

6	C	Gravitational field strength at Earth's surface, $g = \frac{GM}{R_E^2} \quad \text{--- (1)}$ Mean density of Earth, $\rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi R_E^3} \quad \text{--- (2)}$ (2)/(1):

No.	Answer	Solution
		$\frac{\rho}{g} = \frac{3}{4\pi GR_E}$ $\rho = \frac{3}{4\pi \times 6.67 \times 10^{-11} \times 6.37 \times 10^6} \times 9.81$ $= 5512$ $\approx 5510 \text{ kg m}^{-3}$
7	B	Height of pendulum after first two oscillations